KANISH K

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LinkedIn — GitHub — Portfolio

SUMMARY

B.E. Computer Science and Engineering student at Bannari Amman Institute of Technology (graduating 2028, CGPA 7.73/10) focused on Applied Machine Learning, NLP/RAG pipelines, and Full-Stack Backend Systems. Experienced building FastAPI and Express REST APIs, PostgreSQL data models, and evaluated Scikit-learn, PyTorch, and XGBoost pipelines. Smart India Hackathon 2025 Winner and C-DAC Innovation Challenge Runner-up seeking SWE and AI/ML internships.

EDUCATION

B.E. Computer Science and Engineering, Bannari Amman Institute of Technology 2024 – 2028
CGPA: 7.73/10

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C++, C
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks
AI & ML: Scikit-learn, XGBoost, PyTorch, TensorFlow, TF-IDF & NLP, Ollama, ChromaDB, NumPy, Pandas
Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, JWT/Bcrypt, Socket.IO, Pytest
Frontend: React.js, Next.js, TypeScript, Tailwind CSS, Vite, HTML5/CSS3
Databases & Tools: PostgreSQL, Redis, Prisma, SQLAlchemy, Alembic, SQLite, Git, GitHub, Docker, Nginx

PROJECTS

Digital Twin of EHV Substation Monitoring 2025
Smart India Hackathon 2025 – National Winner (Team Oceanberg) | Python, Scikit-learn, TensorFlow, Streamlit, ReportLab

- Co-authored per-component Python diagnostics (Random Forest, Isolation Forest) across transformer, busbar, and breaker modules over simulated SCADA CSV datasets (5,000 training / 6,000 output rows).
- Built the Streamlit CNN-LSTM waveform fault trainer workflow and automated ReportLab PDF / SMTP alerting.

Kiosk Management System (AAZHI) 2024
C-DAC SUVIDHA Innovation Challenge – Runner-up (Team Fork) | React, Express, PostgreSQL, Redis, FastAPI, Scikit-learn

- Contributed multilingual citizen kiosk UI, Aria voice routing (11+ Indian languages), Gas billing/admin/SLA views, and QR mobile tracking fixes across 78 commits on `KANISH-850/aazhi`.
- Integrated with team Express (`pg`, Redis/BullMQ, Socket.IO, Razorpay) and FastAPI ML triage services (LinearSVC spam, TF-IDF duplicates, Random Forest forecasting).

TechSpire – Unified AI Hospital Management System 2026
Sole Author | FastAPI, PostgreSQL 16, SQLAlchemy, Alembic, Ollama (qwen3:4b), ChromaDB, React, Docker

- Consolidated clinical, ward bed, inventory, and reporting modules in a 4-container Docker Compose stack behind Nginx with 14 SQLAlchemy ORM models, Alembic migrations, and HS256 JWT/Bcrypt auth.
- Built on-premise Ollama RAG chat via ChromaDB (SSE streaming) and RandomForestRegressor forecasts on synthetic seeded data (20/20 Pytest tests documented in repository).

Intelligent Multilingual Grievance System 2026
Primary Author | React, TypeScript, Express, Prisma, PostgreSQL, FastAPI, PyTorch, Scikit-learn

- Built a 3-tier civic triage platform supporting 12 Indian languages (LRU-cached translation), explainable TF-IDF + LogisticRegression routing, and multi-issue transactional Prisma dispatch.
- Benchmarked PyTorch MobileNetV3-Small vision triage on 38 held-out real civic images (50.0% test accuracy), pairing confidence flags with human officer review queues.

AquaGuard AI (Neeraran) – Water Resilience ML Audit 2026
ML Audit & Sensor Health Contributor (`kanish` branch) | Python, One-Class SVM, XGBoost, SHAP, FastAPI, Pytest

- Authored comparative ML audit across 46,656 simulated telemetry rows (One-Class SVM 0.7451 F1; XGBoost 100% leak recall, 97.37% precision) and built Layer-1 sensor health decoupling endpoints.

ACHIEVEMENTS

- **Smart India Hackathon 2025:** 1st Place National Grand Finale Winner (Team Oceanberg) — Digital Twin of EHV Substation Monitoring.
- **C-DAC SUVIDHA Innovation Challenge:** National Runner-up (Team Oceanberg) — AAZHI Civic Self-Service Kiosk Platform.